

CHROMIUM CONTENTS OF SOME WIDELY USED INDIGENOUS HYPOGLYCEMIC HERBAL DRUGS

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ABSTRACT

Chromium and other trace elements of some the hypoglycemic plants, *Gymnema sylvestre*, *Momordica charantia*, *Caralluma edulis* and *Berberus lycium*, widely used as herbal drugs by local practitioner were determined by Atomic absorption Spectrophotometry (AAS) and Flameless Atomic Absorption Spectrophotometry.

Chromium contents of these herbal drugs were compared with the reported daily requirements and experimental therapeutic concentration and relationship of chromium contents with their hypoglycemic activity was examined.

INTRODUCTION

Chromium improves glucose tolerance in chromium deficient rats (Mertz W., 1969) has been postulated as an essential part of an Insulin potentiating factor. The beneficial effect of chromium is related to a usable form of chromium and not total chromium. This usable form of chromium may be present in foods, as such or can be biosynthesized within the body by most people (Mertz W et al., 1989). Chromium deficiencies interfere with carbohydrate and lipid metabolism, and severe cases may lead to diabetes mellitus and cardiovascular abnormalities (Khan A. et al., 1990).

Diabetes has been well documented health problem of the western world and has emerged as an important health problem in Pakistan. In Pakistan diabetes is common in age between 41-50 years, in both sexes, with slightly higher figures for females as compared to the males (Samad S.A. 1992).

Different therapeutic measures, however, have advanced to the point where most diabetic patient can expect a nearly normal life-span and productivity, both through dietary control, and through introduction of various oral hypoglycemic agents and variety of insulin preparations isolated from various sources.

Through the ages plants have been used as source of drugs, in cure of diseases (Mertz W., 1969). Many herbs and plants exhibit hypoglycemic activity when taken orally (Nadkarni K.M., 1954; 1984; Mansoor M.Y. et al., 1963; Liwis H.W. & Elvin Lewis M.P.H., 1977). Some of them are widely used by the practitioner of indigenous

systems of medicine either singly or in combination with other plants in sub-continent. An attempt has been made to evaluate the chromium contents and some other trace elements of the hypoglycemic plants *Gymnema sylvestre* (Gurmar), *Momordica charantia* (Karela), *Caralluma edulis* (Choong) and *berberis lyceum* (Kashmal) and a comparison with daily requirement and experimental therapeutic concentration has been made to see if it bears any relationship towards hypoglycemic activity.

MATERIALS AND METHODS

Dry digestion method ($\text{HNO}_3 : \text{HClO}_4$) was employed to prepare extracts of *Gymnema sylvestre* (leaves), *Momordica charantia* (fruits), *caralluma edulis* (Stem) and *Berberis Lycium* (roots).

Instrumental:

Atomic absorption spectroscopic measurements were performed with a Perkin Elmer Model 3300, atomic absorption spectrometer equipped with an HGA600 Graphite furnace, an AS-60 Autosampler and a deuterium background corrector (FAAS).

Reagents:

All the solutions were prepared from analytical reagent grade chemicals using double deionized water. Nitric acid (65%), perchloric acid (70%) and magnesium nitrate (98%) were obtained from E.Merck.

RESULTS AND DISCUSSION

FAAS is one of the recent development in analytical technique, and has been comparatively evaluated and successively used for the determination of chromium, specially in biological material (Ottaway J.M. & Fell G.S., 1986).

An organo-chromium complex, glucose tolerance factor (GTF), synthesized in tissues from absorbed chromium III potentiate the actions of insulin (Robinson C.H. et al., 1985) and ultimately enhance the uptake of glucose by the cells (Uusitupa M.I.J., 1983). Deficiency of this complex either through absolute dietary lack of chromium III or reduced ability to synthesize the specific complex results in glucose intolerance and reduced efficiency of insulin, affecting carbohydrates, amino acid and lipid metabolism. Symptomatic chromium (III) deficiency has been clearly demonstrated in man during prolong intravenous feeding when purified nutrients (amino acid and carbohydrates) do not contain sufficient chromium (III) to meet daily requirements (Ottaway J.M. & Fell 1986). The absorption of trivalent chromium in GTF is about 10 to 25 percent; only 1 percent of inorganic chromium is absorbed (Robinson C.H. et al., 1985).

Supplements of chromium have been found to improve glucose tolerance (Ottaway J.M. & Fell G.S., 1986). In two studies, supplementation of intravenous fluid with 250 microgram (μ g) of chromium per day and 150 microgram (μ g) per day (Ottaway J.M. & Fell G.S., 1986) has reversed severe glucose intolerance, improved nitrogen balance, weight gain and reduced signs of peripheral neuropathy. Hypoglycemic properties of leaves of *Gymnema sylvestre* (Shanmuga Sundaram E.B. et al., 1990; Baskaran K. Ahamat B.K. et al., 1990; Akhtar M.S. 1990 and Dogar I.A. et al., 1990) extract of juice, fruits, seeds of *Momordica charantia* (Liaqat A. et al., 1993; Sirivastaa Y., et al., 1993; Patel K. & Sirivasan K. 1995; Tingda C. et al., 1987; Karunanayake E.H. et al., 1990; Sankaranarynan J. et al., 1993 and William F. et al., 1993) and *Caralluma edulis* (William F. et al., 1993; Akhtar T. et al., 1990; Ali S.I. 1983; Ahmad M.M. & Shaikh M.M. 1989 and Chopra R.N. et al., 1956) have been reported.

Chromium contents are lowest in *Caralluma edulis* and highest in *Gymnema sylvestre* (Table-1). Recommended daily allowances of chromium are 0.05 to 0.2 milligram (mg) per day for adults and 0.01 to 0.12 mg for infants and younger children (1980).

As in genral practice for the treatment of diabetes mellitus these herbal drugs are administrated as such (original plant material) and not in the form of an active component isolated from them. The dispensed drugs are such calculated so as to contain approximately 300-400 microogram (μ g) of chromium per dose. This is an adequate amount to meet with any chromium deficiency and to improve glucose intolerance in non-insulin dependent diabetes mellitus. Most probably the hypoglycemic activity of these plants may be due to their chromium contents.

Besides chromium presence of other metabolically important minerals, Fe, Mn, Cu, Ni, Co and Zn in these plants is also of great significance (Table-1). Occurrence of these elements alongwith chromium may potentiate the hypoglycemic effects or properties of *Gymnema sylvestra* *Momordica charantia*, *Caralluma edulis* and *Berberis lycium*. Obviously role of zinc can not be ignored which is specifically required by insulin.

Table 1
Chromium contents and other trace elements found in various
hypoglycemic plants extract ($\mu\text{g/g}$)

S. No.	Hypoglycemic Plant	Cr	Fe	Mn	Zn	Cu	Ni	Co	Pb
1.	Gymnema Sylvestre	0.19	11.2	3.12	0.26	0.12	0.08	0.02	0.4
2.	Momordica Charantia	0.10	6.1	0.38	0.25	0.08	0.06	0.05	0.3
3.	Caralluma Edulis	0.91	4.5	1.02	0.25	0.04	0.07	0.0	0.05
4.	Berberus Lycium	0.16	17.1	0.64	0.44	0.08	0.58	0.03	0.19

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